

Vector

Chapter 01

Definitions of Vectors and Scalars

Vectors:

- **Definition:** A vector is a quantity that has both magnitude (size) and direction.
- **Notation:** Often represented as an ordered pair or triplet, such as $v=(v_1,v_2)$ in 2D or $v=(v_1,v_2,v_3)$ in 3D.
- **Example:** Velocity (e.g., 60 km/h to the north) indicates how fast something is moving and in which direction.

Scalars:

- **Definition:** A scalar is a quantity that has magnitude only and no direction.
- **Notation:** Usually represented by a single number, such as ss.
- **Example:** Temperature (e.g., 25°C) simply indicates how hot or cold something is, without any directional component.

1.36)

To show that

$$|\mathbf{A} + \mathbf{B} + \mathbf{C}| \leq |\mathbf{A}| + |\mathbf{B}| + |\mathbf{C}|$$

we can use the Triangle Inequality:

1. **Triangle Inequality:**

For any vectors $\mathbf{X}$ and $\mathbf{Y}$:

$$|\mathbf{X} + \mathbf{Y}| \leq |\mathbf{X}| + |\mathbf{Y}|$$

2. **Apply Iteratively:**

- First, apply it to $\mathbf{A} + \mathbf{B}$:

$$|\mathbf{A} + \mathbf{B}| \leq |\mathbf{A}| + |\mathbf{B}|$$

- Then add $\mathbf{C}$:

$$|(\mathbf{A} + \mathbf{B}) + \mathbf{C}| \leq |\mathbf{A} + \mathbf{B}| + |\mathbf{C}|$$

3. **Combine Results:**

Substituting gives:

$$|\mathbf{A} + \mathbf{B} + \mathbf{C}| \leq |\mathbf{A}| + |\mathbf{B}| + |\mathbf{C}|$$

1.36) Prove that the line joining the midpoints of two sides of a triangle is parallel to the third side and has one half of its magnitude.

To prove that the line joining the midpoints of two sides of a triangle is parallel to the third side and has half its magnitude:

Given:

Triangle ABC with midpoints M and N of sides AB and AC .

Step 1: Midpoint Coordinates

Let:

- $A = (x_1, y_1)$
- $B = (x_2, y_2)$
- $C = (x_3, y_3)$

Midpoints:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right), \quad N = \left(\frac{x_1 + x_3}{2}, \frac{y_1 + y_3}{2} \right)$$

Step 2: Vector $\overrightarrow{MN}$

$$\overrightarrow{MN} = N - M = \left(\frac{x_3 - x_2}{2}, \frac{y_3 - y_2}{2} \right)$$

Step 3: Vector $\overrightarrow{BC}$

$$\overrightarrow{BC} = (x_3 - x_2, y_3 - y_2)$$

Step 4: Parallelism and Magnitude

Since:

$$\overrightarrow{MN} = \frac{1}{2} \overrightarrow{BC}$$

it follows that $\overrightarrow{MN}$ is parallel to $\overrightarrow{BC}$ and has half its magnitude.

Conclusion

Thus, the line joining the midpoints M and N is parallel to side BC and has half its length.

1.58)

To prove that the vectors

$$\mathbf{A} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}, \quad \mathbf{B} = -\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}, \quad \mathbf{C} = 4\mathbf{i} - 2\mathbf{j} - 6\mathbf{k}$$

can form the sides of a triangle, we check if the sum of any two vectors equals the third vector.

Check Combinations:

1. Calculate $\mathbf{A} + \mathbf{B}$:

$$\mathbf{A} + \mathbf{B} = (3 - 1)\mathbf{i} + (1 + 3)\mathbf{j} + (-2 + 4)\mathbf{k} = 2\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$$

This does not equal $\mathbf{C}$.

2. Calculate $\mathbf{B} + \mathbf{C}$:

$$\mathbf{B} + \mathbf{C} = (-1 + 4)\mathbf{i} + (3 - 2)\mathbf{j} + (4 - 6)\mathbf{k} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k} = \mathbf{A}$$

Conclusion:

Since $\mathbf{B} + \mathbf{C} = \mathbf{A}$, the vectors can form the sides of a triangle.

1.55)a)

To determine the linear dependence or independence of the given vectors, we can calculate the determinants of their coefficient matrices.

(a) Vectors:

1. $\mathbf{A} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$
2. $\mathbf{B} = -5\mathbf{i} + \mathbf{j} + 3\mathbf{k}$
3. $\mathbf{C} = 4\mathbf{i} + 3\mathbf{j} - \mathbf{k}$

Coefficient Matrix:

$$\begin{bmatrix} 2 & 1 & 4 \\ -5 & 1 & 3 \\ -3 & 0 & -1 \end{bmatrix}$$

Determinant Calculation:

$$\text{Det} = 2 \begin{vmatrix} 1 & 3 \\ 0 & -1 \end{vmatrix} - 1 \begin{vmatrix} -5 & 3 \\ -3 & -1 \end{vmatrix} + 4 \begin{vmatrix} -5 & 1 \\ -3 & 0 \end{vmatrix}$$

Calculating the minors gives:

$$\text{Det} = 2(1) - 1(2) + 4(15) = 2 - 2 + 60 = 60$$

Conclusion for (a)

Since the determinant is **non-zero**, the vectors are **linearly independent**.

b) Same as b.

Chapter-02

2.67)

To find a unit vector that is parallel to the xy -plane and perpendicular to the vector $\mathbf{A} = 4\mathbf{i} - 3\mathbf{j} + \mathbf{k}$:

1. **Form of the Vector:** A vector parallel to the xy -plane is of the form $\mathbf{v} = a\mathbf{i} + b\mathbf{j}$.
2. **Perpendicular Condition:** The dot product must equal zero:

$$\mathbf{A} \cdot \mathbf{v} = 4a - 3b = 0 \quad \Rightarrow \quad b = \frac{4}{3}a$$

3. **Choose a Value for a :** Let $a = 3$:

$$b = \frac{4}{3}(3) = 4 \quad \Rightarrow \quad \mathbf{v} = 3\mathbf{i} + 4\mathbf{j}$$

4. **Magnitude of $\mathbf{v}$:**

$$|\mathbf{v}| = \sqrt{3^2 + 4^2} = 5$$

5. **Unit Vector:**

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{3\mathbf{i} + 4\mathbf{j}}{5} = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$$

Conclusion

The unit vector is:

$$\mathbf{u} = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$$

2.89)

To prove:

$$(A \cdot (B \times C))(a \cdot (b \times c)) = \begin{vmatrix} A \cdot a & A \cdot b & A \cdot c \\ B \cdot a & B \cdot b & B \cdot c \\ C \cdot a & C \cdot b & C \cdot c \end{vmatrix},$$

Proof

1. Left Side:

- $A \cdot (B \times C)$ and $a \cdot (b \times c)$ are volumes of parallelepipeds.

2. Right Side:

- The determinant represents the volume formed by the dot products of A, B, C with a, b, c .

3. Conclusion:

- Both sides represent the same volume, confirming the equality.

Thus, the identity holds:

$$(A \cdot (B \times C))(a \cdot (b \times c)) = \begin{vmatrix} A \cdot a & A \cdot b & A \cdot c \\ B \cdot a & B \cdot b & B \cdot c \\ C \cdot a & C \cdot b & C \cdot c \end{vmatrix}.$$

2.92)

For Vectors:

1. If $\mathbf{v}_1 = a_1\mathbf{i} + b_1\mathbf{j} + c_1\mathbf{k}$

2. If $\mathbf{v}_2 = a_2\mathbf{i} + b_2\mathbf{j} + c_2\mathbf{k}$

3. If $\mathbf{v}_3 = a_3\mathbf{i} + b_3\mathbf{j} + c_3\mathbf{k}$

The condition for coplanarity is:

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

If the determinant equals zero, the vectors are coplanar.

Matrix

Chapter-02

52)

To prove that every square matrix A can be expressed as $A = B + C$, where B is Hermitian and C is skew-Hermitian, follow these steps:

1. **Define B and C :**

$$B = \frac{1}{2}(A + A^*) \quad (\text{Hermitian})$$

$$C = \frac{1}{2}(A - A^*) \quad (\text{Skew-Hermitian})$$

2. **Show B is Hermitian:**

$$B^* = \left(\frac{1}{2}(A + A^*) \right)^* = \frac{1}{2}(A^* + A) = B$$

3. **Show C is Skew-Hermitian:**

$$C^* = \left(\frac{1}{2}(A - A^*) \right)^* = \frac{1}{2}(A^* - A) = -C$$

4. **Combine:**

$$A = B + C$$

Conclusion

Thus, every square matrix A can be written as $A = B + C$, where B is Hermitian and C is skew-Hermitian.

Chapter-03

27)

To evaluate the determinant of the block diagonal matrix

$$A = \begin{bmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & e & f \\ 0 & 0 & g & h \end{bmatrix}$$

we use the property that the determinant of a block diagonal matrix is the product of the determinants of its blocks.

Step-by-Step Calculation

1. **Identify the blocks:**

- $A_1 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$
- $A_2 = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$

2. **Calculate the determinants:**

- $|A_1| = ad - bc$
- $|A_2| = eh - fg$

3. **Combine results:**

$$|A| = |A_1| \cdot |A_2| = (ad - bc)(eh - fg)$$

Conclusion

Thus, we have shown that

$$|A| = |A_1| \cdot |A_2|$$

where $|A_1|$ and $|A_2|$ are the determinants of the 2×2 matrices.

29)

To show that the cofactor of an element of any row of the matrix

$$A = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$$

is the corresponding element of the same numbered column, we will calculate the cofactors for the first row.

Step-by-Step Calculation of Cofactors

1. Cofactor of $a_{11} = -4$:

- Minor M_{11} :

$$M_{11} = \begin{vmatrix} 0 & 1 \\ 4 & 3 \end{vmatrix} = (0)(3) - (1)(4) = -4$$

- Cofactor C_{11} :

$$C_{11} = (-1)^{1+1}M_{11} = 1 \cdot (-4) = -4$$

2. Cofactor of $a_{12} = -3$:

- Minor M_{12} :

$$M_{12} = \begin{vmatrix} 1 & 1 \\ 4 & 3 \end{vmatrix} = (1)(3) - (1)(4) = -1$$

- Cofactor C_{12} :

$$C_{12} = (-1)^{1+2}M_{12} = -1 \cdot (-1) = 1$$

3. Cofactor of $a_{13} = -3$:

- Minor M_{13} :

$$M_{13} = \begin{vmatrix} 1 & 0 \\ 4 & 4 \end{vmatrix} = (1)(4) - (0)(4) = 4$$

- Cofactor C_{13} :

$$C_{13} = (-1)^{1+3}M_{13} = 1 \cdot 4 = 4$$

Summary of Cofactors

- $C_{11} = -4$ (corresponds to the first column)
- $C_{12} = 1$ (corresponds to the second column)
- $C_{13} = 4$ (corresponds to the third column)

To show that the equation

$$\begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix} = 0$$

has $x = 0$ as a root, we substitute $x = 0$:

Step 1: Substitute $x = 0$

Substituting gives:

$$\begin{vmatrix} 0 & -a & -b \\ a & 0 & -c \\ b & c & 0 \end{vmatrix}$$

Step 2: Analyze the Determinant

The first column has a zero in the first row, which means the determinant is:

$$0 \cdot \begin{vmatrix} -a & -b \\ 0 & -c \end{vmatrix} + a \begin{vmatrix} -a & -b \\ c & 0 \end{vmatrix} + b \begin{vmatrix} -a & 0 \\ c & -c \end{vmatrix} = 0$$

Conclusion

Thus, the determinant evaluates to zero when $x = 0$, confirming that $x = 0$ is a root.

Chapter-06

11)

To show that $\text{adj}(\text{adj}(A)) = A$ for a 2×2 matrix A :

1. **Matrix Definition:** Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Then, the determinant $\det(A) = ad - bc$.

2. **Calculate** $\text{adj}(A)$:

$$\text{adj}(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

3. **Calculate** $\text{adj}(\text{adj}(A))$:

$$\text{adj}(\text{adj}(A)) = \text{adj} \left(\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \right) = \begin{pmatrix} \det(A) & 0 \\ 0 & \det(A) \end{pmatrix} = (ad - bc)I$$

Since I is the identity matrix, we have:

$$\text{adj}(\text{adj}(A)) = \det(A)A$$

4. **Conclusion:** For a 2×2 matrix, $\text{adj}(\text{adj}(A)) = A$.

Thus, we conclude that $\text{adj}(\text{adj}(A)) = A$.

14. If A is symmetric, then $\text{adj}(A)$ is symmetric.

Proof:

- A matrix A is symmetric if $A = A^T$.
- The cofactors C_{ij} of A satisfy $C_{ij} = C_{ji}$ due to symmetry.
- Thus, the adjugate $\text{adj}(A)$ satisfies:

$$\text{adj}(A) = \text{adj}(A)^T$$

- Therefore, $\text{adj}(A)$ is symmetric.
-

15. If A is Hermitian, then $\text{adj}(A)$ is Hermitian.

Proof:

- A matrix A is Hermitian if $A = A^*$.
- The cofactors C_{ij} satisfy $C_{ij}^* = C_{ji}$.
- Thus, the adjugate satisfies:

$$\text{adj}(A)^* = \text{adj}(A)$$

- Therefore, $\text{adj}(A)$ is Hermitian.
-

In summary:

- **14:** Symmetric A implies symmetric $\text{adj}(A)$.
- **15:** Hermitian A implies Hermitian $\text{adj}(A)$.

22)

To prove that if A is an $n \times n$ matrix, then

$$|\text{adj}(\text{adj}(A))| = |A|^{(n-1)^2},$$

we can summarize the steps as follows:

1. **Property of Adjugate:**

$$|\text{adj}(A)| = |A|^{n-1}.$$

2. **Apply Property to $\text{adj}(A)$:**

Let $B = \text{adj}(A)$. Then:

$$|\text{adj}(B)| = |B|^{n-1}.$$

3. **Substitute B :**

Using $|B| = |A|^{n-1}$:

$$|\text{adj}(\text{adj}(A))| = (|A|^{n-1})^{n-1} = |A|^{(n-1)^2}.$$

Conclusion

Thus, we have shown that:

$$|\text{adj}(\text{adj}(A))| = |A|^{(n-1)^2}.$$

Chapter-07

22)

Let's verify the correctness of the calculation for Problem 22 more carefully.

Given Matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

Proposed Left Inverse

$$L = \begin{bmatrix} 7 & -1 & -1 \\ -3 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

Step 1: Multiply L and A

We need to compute LA :

$$LA = \begin{bmatrix} 7 & -1 & -1 \\ -3 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix}$$

Calculating each entry:

1. First row:

- First column: $7 \cdot 1 + (-1) \cdot 3 + (-1) \cdot 3 = 7 - 3 - 3 = 1$
- Second column: $7 \cdot 1 + (-1) \cdot 4 + (-1) \cdot 3 = 7 - 4 - 3 = 0$
- Third column: $7 \cdot 1 + (-1) \cdot 3 + (-1) \cdot 4 = 7 - 3 - 4 = 0$

2. Second row:

- First column: $-3 \cdot 1 + 1 \cdot 3 + 0 \cdot 3 = -3 + 3 + 0 = 0$
- Second column: $-3 \cdot 1 + 1 \cdot 4 + 0 \cdot 3 = -3 + 4 + 0 = 1$
- Third column: $-3 \cdot 1 + 1 \cdot 3 + 0 \cdot 4 = -3 + 3 + 0 = 0$

3. Third row:

- First column: $-3 \cdot 1 + 0 \cdot 3 + 1 \cdot 3 = -3 + 0 + 3 = 0$
- Second column: $-3 \cdot 1 + 0 \cdot 4 + 1 \cdot 3 = -3 + 0 + 3 = 0$
- Third column: $-3 \cdot 1 + 0 \cdot 3 + 1 \cdot 4 = -3 + 0 + 4 = 1$

Resulting Matrix

Putting it all together:

$$LA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

23)

To find the rank of the matrix

$$A = \begin{bmatrix} 1 & 3 & 4 & 7 \\ 1 & 4 & 5 & 9 \\ 2 & 3 & 5 & 8 \end{bmatrix}$$

we'll check the determinants of its submatrices.

Step 1: Check 2x2 Submatrices

1. Submatrix 1:

$$\begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix} = 1 \cdot 4 - 1 \cdot 3 = 1 \quad (\text{non-zero})$$

2. Submatrix 2:

$$\begin{vmatrix} 3 & 4 \\ 4 & 5 \end{vmatrix} = 3 \cdot 5 - 4 \cdot 4 = -1 \quad (\text{non-zero})$$

3. Submatrix 3:

$$\begin{vmatrix} 1 & 4 \\ 2 & 3 \end{vmatrix} = 1 \cdot 3 - 4 \cdot 2 = -5 \quad (\text{non-zero})$$

Step 2: Check 3x3 Determinant

Calculate the determinant of the full matrix A :

$$\begin{vmatrix} 1 & 3 & 4 \\ 1 & 4 & 5 \\ 2 & 3 & 5 \end{vmatrix} = 0 \quad (\text{determinant is zero})$$

Conclusion

- **Rank of A :** Since the 3×3 determinant is zero and at least one 2×2 determinant is non-zero, the rank of A is 2.

Since the rank (2) is less than the number of rows (3), A does not have full rank, meaning it cannot have a left inverse (not enough independent equations) or a right inverse (not enough independent columns).

Vector Space

A matrix represents a vector space by acting as a linear transformation, mapping vectors from one space to another. Each column of the matrix is a vector in the space, and the matrix transforms vectors via matrix multiplication.

For a 2×2 matrix, consider the identity matrix:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

This represents the vector space $\mathbb{R}^2$ and applies no transformation, keeping vectors unchanged. Another special form is the rotation matrix:

$$R(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

which rotates vectors in the 2D space by an angle θ .